SWARUP KUMAR SUBUDHI

PhD Candidate, Mechanical Engineering · R&D & Advanced Manufacturing Engineer

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PROFESSIONAL SUMMARY

PhD candidate (Mechanical Engineering, UMD) with 4+ years of industry leadership at Tata Motors and 3+ years of sponsored research in printed electronics, additive manufacturing, and soft magnetic materials. Proven track record of translating fundamental research into testable, qualified hardware — from ±50 µm silver-ink interconnects to Six Sigma warranty-cost reductions. Deep expertise in reliability engineering, materials characterization, and multi-physics FEA. Authorized to work in the US on STEM OPT (E-Verify); available from mid-2026.

EXPERIENCE

Graduate Research Assistant · *University of Maryland, College Park* Jan 2022 – Present

Fine-Line Printing of Silver - (Additive Manufacturing)

- Achieved ±10 µm print tolerances for high-density (50-100 µm) silver-ink circuits on space-grade substrates, enabling high-precision printed electronics qualified for launch environments
- Conducted vibration qualification tests (launch-load profiles) and environmental reliability testing on printed specimens; established pass/fail criteria and documented qualification matrix
- Developed innovative extrusion-printing toolpaths for irregular substrate geometries; reduced print defects by iterative ink-rheology tuning (viscosity, surface tension, particle size distribution)
- Authored feasibility studies and test documentation for program reviews

Thermal Cycling Reliability of Printed Electronics

- Built multi-layer FEA models (ANSYS/Abaqus) of printed dielectric and conductive stacks subjected to −40 °C to +140 °C thermal cycling; identified failure loci via inelastic strain accumulation and creep in sintered silver
- Correlated FEA predictions with experimental SEM and cross-section microscopy failure analysis

RF Printed Electronics Reliability

- Fabricated MIS capacitors and coplanar waveguides by direct-write extrusion printing of dielectric (NEA121) and silver pastes on curved substrates
- Characterized RF performance (S-parameters, reflection coefficients, efficiency) and qualified specimens against program specs

Doctoral Research — Magnetic Drops & Soft Magnetic Nanocomposites

- Demonstrated magnetohydrodynamic de-encapsulation of superparamagnetic nanoparticles (MNPs) from liquid drops and cargo transfer between droplets — enabling next-generation lab-on-chip drug delivery
- Fabricated flexible PDMS-based soft magnetic films (biocompatible polymer + Fe_3O_4 MNPs) for RF shielding and magnetic damping; published in Flexible and Printed Electronics (2024)
- Developed Python/CV2 droplet and aggregate tracking algorithms for high-frame-rate video analysis

Senior Manager, Manufacturing — Transmissions · *Tata Motors Ltd. CVBU, India* 2019 – 2020

- Reduced gearbox bearing warranty failures by ~50% for 24-MIS vehicles through a cross-functional Six Sigma project (NPI, QA, SCM, R&D); realized annual savings of ₹4.53 Cr for the LKP2518 tipper family
- Improved Gearbox Assembly Line OLE by ~12% by re-balancing takt time at bottleneck stations and deploying PFMEA-driven quality controls; used RULA ergonomics assessments to eliminate operator-related defect sources
- Implemented IoT-based low-cost ANDON monitoring system to capture real-time assembly line losses; deployed Ishikawa-driven corrective action plans
- Led a cross-functional team of 3–5 engineers across NPI, QA, SCM, and R&D divisions

Assistant Manager, NPI — Transmissions · *Tata Motors Ltd. CVBU, India* 2016 – 2019

- Launched auxiliary transmission (gearbox) system for the TATA Kestrel 8×8 defense vehicle through proto stage; managed full NPI process including DFA/DFM, part qualification, PLM integration, and test matrix development

- Delivered 100% on-time production adherence for 6×6 and 8×8 defense truck gearboxes across 10 new product launches
- Executed value engineering case studies on synchronizing parts and hydraulic tipping kits, generating measurable cost savings

Graduate Teaching Assistant · *University of Maryland, College Park* Jan 2021 – Dec 2021

- Instructed lab and studio sections for Fluid Mechanics (ENME331) and Engineering Mechanics I (ENES102); held weekly office hours for 80+ students

EDUCATION

Ph.D., Mechanical Engineering · University of Maryland, College Park, MD Expected mid-2026
Dissertation: Magnetic Films & Drops — Interactions with Viscous Mediums; *GPA:* 3.83; *Key courses:* Probabilistic PoF & Accelerated Testing, Failure Mechanisms, Composite Materials, Interfacial Fluid Mechanics

M.S., Mechanical Engineering · University of Maryland, College Park, MD 2021

B.Tech., Mechanical Engineering · National Institute of Technology, Rourkela, India 2016

TECHNICAL SKILLS

Additive Mfg & Fabrication: Extrusion/Direct-Write Printing, FDM/SLA 3D Printing, Subtractive CNC Machining, Laser Tooling, Mold Development, nScryptPro, Autodesk 360
CAD / CAE / CFD: SolidWorks, Creo (Pro-E), ANSYS Mechanical/Workbench/Icepak, Abaqus CAE, COMSOL Multiphysics, Fluent
Reliability & Testing: Thermal Cycling, Temp-Humidity Bias (THB), Vibration/Shock Loading, Failure Analysis, Physics of Failure (PoF), Acceleration Factors, Weibull Analysis
Materials Characterization: SEM, TMA/CTE, DMA, DSC, TGA, Optical/Laser Microscopy, Surface Profiling, GD&T, MSA / Gauge R&R
Rheology: Viscosity, Surface Tension, Zeta-Potential, Particle Size Distribution — applied to printable inks
Statistical & Quality: Six Sigma, DoE, ANOVA, SPC (Cpk), PFMEA, DFA/DFM, NPI Process, PLM, BOM
Computation: Python (CV2, image processing, point tracking), MATLAB/Simulink (optimization, image processing), R, C/C++, HTML5/CSS3/JS
Instrumentation & Control: Arduino, Python/MATLAB GUI, sensor integration, circuit design, prototyping

RELEVANT PUBLICATIONS

- Subudhi, S.K., Zhao, B., Wang, X., Ting, J., Takeuchi, I., Dasgupta, A., and Das, S. (2024). Flexible and twistable free-standing PDMS–magnetic-nanoparticle-based soft magnetic films with robust magnetic properties. Flexible and Printed Electronics, 9(1), 015013.
- Subudhi, S.K., Chandel, G.R., Sivasankar, V., and Das, S. (2024). Magnetic nanoparticle aggregation and complete de-encapsulation from a liquid drop interior. ACS Applied Materials & Interfaces, 16(46), 64253–64263.
- Subudhi, S.K. and Das, S. (2023). Reliability of lab-on-a-chip technologies for wearable electronics: a perspective. Frontiers in Sensors, 4, 1283402.
- Zhao, B., Sivasankar, V.S., Subudhi, S.K., et al. (2023). Printed CNT-based humidity sensors on surfaces of widely varying curvatures. ACS Applied Nano Materials, 6(2), 1459–1474.
- Zhao, B., Sivasankar, V., Subudhi, S.K., et al. (2023). Three-dimensional deposits from drying particle-laden drops. Langmuir, 39(28), 9773–9784.

WORK AUTHORIZATION

F-1 visa holder; eligible for 12-month OPT + 24-month STEM OPT extension (36 months total work authorization). Employer requires E-Verify enrollment only — no H-1B petition or attorney fees required for initial employment period. Available to start mid-2026.